

VISEU, Portugal

WMO: 085600

Lat: 40.7150N

Lon: 7.8958W

Elev: 2115

StdP: 13.61

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.4	33.5	16.5	13.7	41.1	21.1	17.1	41.7	24.7	39.3	22.5	41.0	11.3	70	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.9	91.5	65.8	88.0	64.8	84.9	64.0	68.9	84.0	67.5	82.3	66.1	80.3	6.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	97.3	73.3	62.7	92.2	71.4	61.4	88.1	69.8	34.2	83.9	33.0	82.7	31.9	80.5	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.2	19.3	17.0		DB	27.9	97.1	2.8	2.9	25.8	99.2	24.2	100.9	22.6	102.5	20.5	104.7
				WB	25.0	72.1	3.0	1.9	22.9	73.5	21.2	74.6	19.5	75.7	17.4	77.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	44.9	46.5	50.2	52.7	58.4	65.1	69.2	70.5	66.3	58.8	49.5	46.1
	DBStd	11.11	4.78	5.76	6.44	7.26	7.36	7.41	7.02	6.78	6.64	6.70	5.43	4.50
	HDD50	646	167	127	80	49	8	0	0	0	0	7	74	134
	HDD65	3680	622	519	461	376	231	90	35	19	62	214	466	585
	CDD50	3050	9	28	85	129	269	454	596	636	490	281	58	15
	CDD65	609	0	0	0	7	27	93	166	190	102	24	0	0
	CDH74	5625	0	0	3	37	236	897	1659	1873	792	128	0	0
	CDH80	2032	0	0	0	0	48	293	664	756	243	28	0	0
Wind	WSAvg	7.8	7.8	8.4	8.7	8.2	7.6	7.1	6.9	7.3	7.3	7.7	8.2	8.3
Precipitation	PrecAvg	43.3	6.5	4.2	3.6	4.1	3.2	1.2	0.6	0.8	2.3	5.7	5.4	5.8
	PrecMax	63.5	17.7	13.3	15.8	11.9	8.7	4.0	2.3	3.7	6.3	12.2	16.2	21.2
	PrecMin	25.5	0.0	0.0	0.0	1.2	0.1	0.0	0.0	0.1	0.1	0.9	0.0	0.2
	PrecStd	11.1	4.8	3.2	4.1	2.9	2.3	1.0	0.6	0.9	1.9	3.1	3.9	5.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	66.0	72.8	78.3	85.4	91.5	95.1	96.3	91.3	84.0	67.8	61.1
		MCWB	50.4	52.1	53.2	58.7	63.6	66.2	66.3	65.8	65.0	58.5	54.8	51.9
	2%	DB	56.9	62.1	68.6	73.8	79.7	86.8	91.1	91.8	86.2	77.2	63.3	57.1
		MCWB	48.2	50.2	51.8	56.7	61.5	65.1	66.1	65.5	62.5	58.7	54.4	49.9
	5%	DB	53.7	58.5	65.1	70.0	76.1	83.4	87.3	87.9	82.4	72.6	59.7	54.6
		MCWB	48.3	48.8	50.8	55.3	59.6	64.3	65.3	64.5	61.4	57.6	53.6	49.8
	10%	DB	51.5	55.0	61.2	65.2	72.1	79.3	83.4	84.2	78.6	68.8	57.1	52.6
		MCWB	48.2	47.7	49.6	53.4	58.0	63.4	64.9	63.9	60.6	56.5	52.3	49.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	54.7	56.3	61.1	66.4	70.6	70.8	70.4	67.2	64.1	59.9	55.1
		MCDB	55.7	61.2	65.1	73.7	80.6	83.1	85.5	86.8	83.9	72.0	61.9	56.7
	2%	WB	51.9	52.6	54.4	58.5	63.6	68.2	68.9	68.3	65.4	62.3	57.3	53.1
		MCDB	53.2	57.7	62.5	70.3	75.4	81.6	84.3	84.0	79.6	69.1	60.5	54.7
	5%	WB	50.5	50.8	52.8	56.3	61.4	66.4	67.5	66.8	63.8	60.7	55.4	51.5
		MCDB	51.9	55.2	60.2	66.9	72.9	79.1	82.6	82.2	76.6	66.4	58.0	53.0
	10%	WB	49.0	49.1	51.2	54.1	59.5	64.6	65.9	65.5	62.3	59.1	53.5	49.9
		MCDB	50.9	53.3	58.0	63.3	69.7	76.9	79.9	80.1	73.6	64.8	55.7	51.6
Mean Daily Temperature Range	5% DB	MDBR	11.0	13.3	14.7	15.5	17.8	20.2	22.9	22.9	19.1	13.7	11.2	10.7
		MCDBR	14.2	17.2	19.8	21.4	22.7	24.4	25.8	25.9	23.4	19.5	14.3	13.0
		MCWBR	9.4	9.7	10.4	10.5	11.5	11.9	11.2	10.4	9.6	9.0	8.7	8.8
	5% WB	MCDBR	9.8	14.1	16.1	19.8	21.0	23.1	25.1	24.2	21.0	14.3	10.8	9.8
		MCWBR	7.8	8.9	9.4	10.5	11.3	12.2	11.9	11.0	9.6	8.0	8.0	8.0
Clear-Sky Solar Irradiance	taub	0.298	0.310	0.341	0.344	0.351	0.354	0.344	0.358	0.346	0.334	0.308	0.296	
	taud	2.549	2.508	2.431	2.427	2.427	2.431	2.441	2.396	2.459	2.517	2.548	2.553	
	Ebn at Noon	277	289	289	295	293	291	293	286	284	276	270	268	
	Edh at Noon	23	27	33	35	36	36	35	36	31	27	23	21	
All-Sky Solar Radiation	RadAvg	614	934	1271	1631	2015	2271	2371	2115	1623	1040	683	553	
	RadStd	98	135	188	203	207	141	92	100	122	124	121	82	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (16 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90

Nomenclature: See separate page